

Module 7

Financial Management

Answers

The suggested answers are longer than what candidates are expected to give in the examination. The purpose of the suggested answers is meant to help candidates in their revision and learning. The suggested answers may not contain all the correct points and candidates should note that credit will be awarded for valid answers which may not fully covered in the suggested answers.

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer 1(a)(i)

$$\begin{aligned}\text{Bond price} &= \text{HK\$25 (PVIFA 4\%, 20)} + \text{HK\$1,000 (PVIF 4\%, 20)} \\ &= \text{HK\$796.15}\end{aligned}$$

$$\begin{aligned}\text{Number of bonds issued} &= \text{HK\$50,000,000} / \text{HK\$796.15} \\ &= 62,803 \text{ (round up to the nearest integer)}\end{aligned}$$

Answer 1(a)(ii)

$$\begin{aligned}\text{Bond price} &= \text{HK\$1,000 (PVIF 4\%, 20)} \\ &= \text{HK\$456.39}\end{aligned}$$

$$\begin{aligned}\text{Number of bonds issued} &= \text{HK\$50,000,000} / \text{HK\$456.39} \\ &= 109,556 \text{ (round up to the nearest integer)}\end{aligned}$$

Answer 1(b)(i)

$$\begin{aligned}\text{Repayment upon maturity} &= 62,803 \times (\text{HK\$1,000} + \text{HK\$25}) \\ &= \text{HK\$64,373,075}\end{aligned}$$

Answer 1(b)(ii)

$$\begin{aligned}\text{Repayment upon maturity} &= 109,556 \times \text{HK\$1,000} \\ &= \text{HK\$109,556,000}\end{aligned}$$

Answer 1(c)

The bond without coupon payment can be referred as "zero coupon bond" or simply "zeroes".

Bondholders of the zero coupon bond could pay a much lesser amount, or at a deep discount, to buy the bond investment at the expense of receiving no interest payment over the life of bond issuing period until the retirement of the bond. Upon maturity, bondholders can get back the whole principal amount to arrive their investment goal.

From the issuer's perspective, the company issuing the zero coupon bond is not obliged to pay interest during the life of this bond; thus, it alleviates the pressure of maintaining cash level for regular bond interest payment.

However, the issuer of the zero coupon bond will repay the total interest payments over the ten years together with the principal amount in one go upon the maturity of the bond. As such, the repayment amounts calculated in Questions 1(b)(i) and 1(b)(ii) are different.

Answer 1(d)

- Debt financing is usually cheaper than equity financing.
- Debt financing has tax advantage. Companies could borrow to a point when the tax benefit is offset by the cost of financial distress.
- Equity financing will dilute existing shareholders control. As such, equity financing is usually not a top priority.

Answer 2(a)

The key risk indicator ("KRI") is used to monitor known risks and to mitigate their impact. The purpose of the KRI is to provide early notice to management of risk exposure and enable a company to measure and anticipate risk.

Process in developing a KRI program is as follows:

1. Identification of major risks.
2. Listing the underlying causes.
3. Analysis of the metrics in these causes.
4. Defining the risk appetite (thresholds and gaps in metrics).
5. Describing relevant data, frequency of events, and reporting systems.
6. Backtesting to determine how experience could prevent future occurrences.

This is a circular process that concludes by returning to the beginning and again monitoring major risks.

Answer 2(b)

YMCL is exposed to exchange rate risk for its foreign sales revenue generated from the UK, and interest rate risk arising from the fluctuating interest rate movement for the outstanding Japanese Yen loan.

YMCL also faces foreign investment risk, such as whether the existing trade environment and regulatory requirements in the UK, as well as existing custom duties and procedures for YMCL's export to the UK could be maintained after the Brexit.

Answer 2(c)

To alleviate the risk arising from foreign sales revenue generated from exports to the UK, YMCL may consider entering into a currency forward agreement to hedge against the variation in British Pound. By buying a currency forward agreement, YMCL can lock in the exchange rate in converting the sales turnover revenue, net of the cost of goods sold for the mentioned manufacturing parts sourced from the UK, back into local currency of Japanese Yen.

Answer 2(d)

YMCL can alleviate the interest rate risk arising from the fluctuating interest rate on the outstanding JPY200 billion loan by making use of the derivative, such as an interest rate swap.

An interest rate swap is a forward contract involving two parties exchanging interest payments on the underlying debts. It usually involves the exchange of a fixed interest rate for a floating rate, or vice versa, or a mixture of multiple swap transactions in order to reduce exposure to fluctuations in interest rates.

By employing a swap, YMCL can better estimate the expected interest expenses incurred on the outstanding loan.

Answer 2(e)

There are four financial risk management approaches as listed below:

1. Risk avoidance
2. Risk mitigation
3. Risk transfer
4. Risk acceptance

Answer 3(a)

Key performance indicators ("KPIs") are values that can be identified, measured and used to track the ability of a company to implement strategy successfully and achieve its mission.

The key features of the effective set of KPIs used for the performance measurement are:

1. Simple and easy to understand. Everyone in the company must have a clear idea of what the KPIs entail.
2. Readily quantifiable. KPIs should be easy to measure.

3. Easy to monitor. Performance evaluation requires measurement of KPIs at regular time intervals.
4. Linked to strategy. KPIs must bear direct link to company strategy.

Answer 3(b)

In HK\$' million	<u>Cookies</u>	<u>Chocolates</u>	<u>Ice-creams</u>
Net profit	7,000	3,500	2,900
Investment in total assets	36,000	12,000	18,000
Return on investment ("ROI") (Net profit/ Total assets)	19.44%	29.17%	16.11%
Required rate of return	12.0%	18.0%	23.0%
Required return	4,320	2,160	4,140
Residual income ("RI") (Net profit – Required return)	2,680	1,340	-1,240

Answer 3(c)

From the ROI results, Chocolates Division is the most well performed division among the three divisions.

On the other hand, Cookies Division got the highest RI among the three divisions.

Ice-creams Division is the worst performing division among the three divisions resulting low ROI and negative RI.

Using RI in terms of absolute dollar amount is more consistent with the corporate goal to maximise the total profitability.

Answer 3(d)

The non-financial measures for the performance evaluation are listed as below:

1. Customer satisfaction and retention.
2. Suppliers' trust.
3. Percentage of deliveries not made on time.
4. Number of new products.
5. Employee turnover rate.
6. Employee satisfaction index.

Answer 4(a)

Expected returns for each stock:

$$E(R_A) = 0.6(15\%) + 0.4(4\%) = 10.60\%$$

$$E(R_B) = 0.6(23\%) + 0.4(-19\%) = 6.20\%$$

$$E(R_C) = 0.6(-8\%) + 0.4(16\%) = 1.60\%$$

Portfolio returns under each state of economy:

$$(R_B) = 0.3(15\%) + 0.3(23\%) + 0.4(-8\%) = 8.20\%$$

$$(R_R) = 0.3(4\%) + 0.3(-19\%) + 0.4(16\%) = 1.90\%$$

Expected return for the portfolio:

$$E(R_P) = 0.6(8.2\%) + 0.4(1.9\%) = 5.68\%$$

Answer 4(b)

Stock A:

$$\text{Variance} = 0.6 (0.15 - 0.106)^2 + 0.4 (0.04 - 0.106)^2 = 0.002904$$

$$\text{Standard deviation} = 5.39\%$$

Stock B:

$$\text{Variance} = 0.6 (0.23 - 0.062)^2 + 0.4 (-0.19 - 0.062)^2 = 0.042336$$

$$\text{Standard deviation} = 20.58\%$$

Stock C:

$$\text{Variance} = 0.6 (-0.08 - 0.016)^2 + 0.4 (0.16 - 0.016)^2 = 0.013824$$

$$\text{Standard deviation} = 11.76\%$$

Portfolio:

$$\text{Variance} = 0.6(0.082 - 0.0568)^2 + 0.4(0.019 - 0.0568)^2 = 0.000953$$

$$\text{Standard deviation} = 3.09\%$$

Answer 4(c)

Portfolio standard deviation using the weighted average of individual stock's standard deviation:

$$0.3 (5.39\%) + 0.3 (20.58\%) + 0.4 (11.76\%) = 12.50\%$$

The expected return of a portfolio is the weighted average of each individual asset's return. On the other hand, the standard deviation of a portfolio should not be the weighted average of the standard deviation of each individual asset.

Portfolio standard deviation computed in Question 4(b) is much smaller than the portfolio standard deviation using the weighted average of individual stock's standard deviation computed above. This result clearly shows that risk can be substantially reduced without an equivalent reduction in the expected return due to the effect of diversification according to the systematic risk principle.

The risk reduced refers to unique/ unsystematic/ diversifiable risk, but not the systematic/ market risk, as the three stocks move in different directions and are uncorrelated with each other.

Answer 4(d)

Stock B has the highest total risks because it has the highest standard deviation.

According to systematic risk principle, Stock C should be rewarded with the highest required return because it has the highest beta reflecting the systematic risk.

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